

A passive cartpole in MuJoCo

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The cartpole is the simplest physics-engine model that still looks like something: a cart slides on a frictionless rail with a thin pole hinged on top. With no controller, released from a small offset, the pole should topple under gravity while the cart barely moves, since the rail absorbs almost all of the hinge's reaction force. We ran a four-second passive simulation and recorded the fall.

Methods

The model is a small MJCF cartpole:

- one slide joint for the cart along the x-axis
- one hinge joint for the pole about the y-axis
- damping coefficients 0.05 and 0.005 for the cart and pole
- timestep 0.005 s, simulation length 4 s
- initial pole offset $\theta_0 = 0.15$ rad ($\approx 8.6^\circ$); everything else starts at rest

The simulation is stepped and a frame sampled every 1/60 s into the video below.

Results

[Video — Passive cartpole released from a small offset of $\theta_0 = 0.15$ rad: with no controller the pole topples under gravity while the cart stays almost stationary on the rail. · view the web edition to play.]

The pole crosses 60° from vertical at $t \approx 0.6$ s, ends the run essentially horizontal (final angle 89.5°), and the cart barely moves, only $\approx 3.1 \times 10^{-5}$ m of recoil over the full four seconds, because almost all the angular momentum of the falling pole is absorbed by the rail's reaction force rather than by translating the cart.

Run parameters

Parameter	Value
theta0	0.15
duration	4
fps	60
fall_time_s	0.595
final_angle_deg	89.52964154260019
max_cart_displacement_m	0.00003051021168075732